Question ID 548a4929

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 548a4929

The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the *xy*-plane has an *x*-intercept at $(a, 0)$ and a *y*-intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$?

A. 21

B. 28

C. 32

D. 35

ID: 548a4929 Answer

Correct Answer:
A

Rationale

Choice A is correct. The *x*-intercept of a graph in the *xy*-plane is the point on the graph where $y = 0$. It's given that function h is defined by $h(x) = 4x + 28$. Therefore, the equation representing the graph of $y = h(x)$ is $y = 4x + 28$. Substituting 0 for y in the equation $y = 4x + 28$ yields $0 = 4x + 28$. Subtracting 28 from both sides of this equation yields $-28 = 4x$. Dividing both sides of this equation by 4 yields $-7 = x$. Therefore, the *x*-intercept of the graph of $y = h(x)$ in the *xy*-plane is $(-7, 0)$. It's given that the *x*-intercept of the graph of $y = h(x)$ is $(a, 0)$. Therefore, $a = -7$. The *y*-intercept of a graph in the *xy*-plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $y = 4x + 28$ yields $y = 4(0) + 28$, or $y = 28$. Therefore, the *y*-intercept of the graph of $y = h(x)$ in the *xy*-plane is $(0, 28)$. It's given that the *y*-intercept of the graph of $y = h(x)$ is $(0, b)$. Therefore, $b = 28$. If $a = -7$ and $b = 28$, then the value of $a + b$ is $-7 + 28$, or 21.

Choice B is incorrect. This is the value of b, not $a + b$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $-a + b$, not $a + b$.

Question Difficulty:
Medium

Question ID e62cfe5f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: e62cfe5f

According to a model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to four times the body weight of the bee, in grams. According to the model, what would be the head width, in millimeters, of a worker bumblebee that has a body weight of 0.5 grams?

ID: e62cfe5f Answer

Rationale

The correct answer is 2.6. According to the model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to 4 times the body weight, in grams, of the bee. Let x represent the body weight, in grams, of a worker bumblebee and let y represent the head width, in millimeters. Translating the verbal description of the model into an equation yields $y = 0.6 + 4x$. Substituting 0.5 grams for x in this equation yields $y = 0.6 + 4(0.5)$, or $y = 2.6$. Therefore, a worker bumblebee with a body weight of 0.5 grams has an estimated head width of 2.6 millimeters. Note that 2.6 and 13/5 are examples of ways to enter a correct answer.

Question Difficulty:
Medium

Question ID e3cf671f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: e3cf671f

The function f is defined by $f(x) = 4x + k(x - 1)$, where k is a constant, and $f(5) = 32$. What is the value of $f(10)$?

ID: e3cf671f Answer

Correct Answer:
67

Rationale

The correct answer is 67. It's given that $f(5) = 32$. Therefore, for the given function f, when $x = 5$, $f(x) = 32$. Substituting 5 for x and 32 for $f(x)$ in the given function $f(x) = 4x + k(x - 1)$ yields $32 = 4(5) + k(5 - 1)$, or $32 = 20 + 4k$. Subtracting 20 from each side of this equation yields $12 = 4k$. Dividing each side of this equation by 4 yields $k = 3$. Substituting 3 for k in the given function $f(x) = 4x + k(x - 1)$ yields $f(x) = 4x + 3(x - 1)$, which is equivalent to $f(x) = 4x + 3x - 3$, or $f(x) = 7x - 3$. Substituting 10 for x into this equation yields $f(10) = 7(10) - 3$, or $f(10) = 67$. Therefore, the value of $f(10)$ is 67.

Question Difficulty:
Medium

Question ID a5834ea4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a5834ea4

$$f(x) = 39$$

For the given linear function f, which table gives three values of x and their corresponding values of $f(x)$?

A.

x	$f(x)$
0	0
1	0
2	0

B.

x	$f(x)$
0	39
1	39
2	39

C.

x	$f(x)$
0	0
1	39
2	78

D.

x	$f(x)$
0	39
1	0
2	-39

ID: a5834ea4 Answer

Correct Answer:

B

Rationale

Choice B is correct. For the given linear function f, $f(x)$ must equal 39 for all values of x. Of the given choices, only choice B gives three values of x and their corresponding values of $f(x)$ for the given linear function f.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:
Medium

Question ID 7e3f8363

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 7e3f8363

In the *xy*-plane, the graph of the linear function f contains the points $(0, 3)$ and $(7, 31)$. Which equation defines f, where $y = f(x)$?

A. $f(x) = 28x + 34$

B. $f(x) = 3x + 38$

C. $f(x) = 4x + 3$

D. $f(x) = 7x + 3$

ID: 7e3f8363 Answer

Correct Answer:
C

Rationale

Choice C is correct. In the *xy*-plane, an equation of the graph of a linear function can be written in the form $f(x) = mx + b$, where m represents the slope and $(0, b)$ represents the *y*-intercept of the graph of $y = f(x)$. It's given that the graph of the linear function f, where $y = f(x)$, in the *xy*-plane contains the point $(0, 3)$. Thus, $b = 3$. The slope of the graph of a line containing any two points (x_1, y_1) and (x_2, y_2) can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $(0, 3)$ and $(7, 31)$, it follows that the slope of the graph of the line containing these points is $m = \frac{31-3}{7-0}$, or $m = 4$. Substituting 4 for m and 3 for b in $f(x) = mx + b$ yields $f(x) = 4x + 3$.

Choice A is incorrect. This function represents a graph with a slope of 28 and a *y*-intercept of $(0, 34)$.

Choice B is incorrect. This function represents a graph with a slope of 3 and a *y*-intercept of $(0, 38)$.

Choice D is incorrect. This function represents a graph with a slope of 7 and a *y*-intercept of $(0, 3)$.

Question Difficulty:
Medium

Question ID c1bd5301

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c1bd5301

A model predicts that a certain animal weighed 241 pounds when it was born and that the animal gained 3 pounds per day in its first year of life. This model is defined by an equation in the form $f(x) = a + bx$, where $f(x)$ is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. What is the value of a?

ID: c1bd5301 Answer

Correct Answer:
241

Rationale

The correct answer is 241. For a certain animal, it's given that a model predicts the animal weighed 241 pounds when it was born and gained 3 pounds per day in its first year of life. It's also given that this model is defined by an equation in the form $f(x) = a + bx$, where $f(x)$ is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. It follows that a represents the predicted weight, in pounds, of the animal when it was born and b represents the predicted rate of weight gain, in pounds per day, in its first year of life. Thus, the value of a is 241.

Question Difficulty:
Medium

Question ID 620fe971

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 620fe971

A team of workers has been moving cargo off of a ship. The equation below models the approximate number of tons of cargo, y, that remains to be moved x hours after the team started working.

$$y = 120 - 25x$$

The graph of this equation in the xy-plane is a line. What is the best interpretation of the x-intercept in this context?

A. The team will have moved all the cargo in about 4.8 hours.

B. The team has been moving about 4.8 tons of cargo per hour.

C. The team has been moving about 25 tons of cargo per hour.

D. The team started with 120 tons of cargo to move.

ID: 620fe971 Answer

Correct Answer:
A

Rationale

Choice A is correct. The x-intercept of the line with equation $y = 120 - 25x$ can be found by substituting 0 for y and finding the value of x. When $y = 0$, $x = 4.8$, so the x-intercept is at (4.8, 0). Since y represents the number of tons of cargo remaining to be moved x hours after the team started working, it follows that the x-intercept refers to the team having no cargo remaining to be moved after 4.8 hours. In other words, the team will have moved all of the cargo after about 4.8 hours.

Choice B is incorrect and may result from incorrectly interpreting the value 4.8. Choices C and D are incorrect and may result from misunderstanding the x-intercept. These statements are accurate but not directly relevant to the x-intercept.

Question Difficulty:
Medium

Question ID 17d80dc3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 17d80dc3

In the *xy*-plane, line k has a slope of 5 and a *y*-intercept of $(0, -35)$. What is the *x*-coordinate of the *x*-intercept of line k?

ID: 17d80dc3 Answer

Correct Answer:
7

Rationale

The correct answer is 7. An equation of a line in the *xy*-plane can be written in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the *y*-intercept of the line. It's given that line k has a slope of 5 and a *y*-intercept of $(0, -35)$. Therefore, $m = 5$ and $b = -35$. Substituting 5 for m and -35 for b in the equation $y = mx + b$ yields $y = 5x - 35$. The *x*-intercept of a line in the *xy*-plane is the point where the line intersects the *x*-axis, which is a point with a *y*-coordinate of 0. Substituting 0 for y in the equation $y = 5x - 35$ yields $0 = 5x - 35$. Adding 35 to both sides of this equation yields $35 = 5x$. Dividing both sides of this equation by 5 yields $7 = x$. Therefore, the *x*-coordinate of the *x*-intercept of line k is 7.

Question Difficulty:
Medium

Question ID 1c29bfd1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1c29bfd1

The pressure exerted on a scuba diver at sea level is 14.70 pounds per square inch (psi). For each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by 0.44 psi. What is the total pressure, in psi, exerted on the scuba diver at 105 feet below sea level?

A. 60.90

B. 31.50

C. 14.70

D. 0.44

ID: 1c29bfd1 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that the pressure exerted on a scuba diver at sea level is 14.70 pounds per square inch (psi). It's also given that for each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by 0.44 psi. The total pressure, in psi, exerted on the scuba diver at x feet below sea level can be represented by the expression $0.44x + 14.70$. Substituting 105 for x in this expression yields $0.44(105) + 14.70$, or 60.90. Therefore, the total pressure exerted on the scuba diver at 105 feet below sea level is 60.90 psi.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pressure, in psi, exerted on the scuba diver at sea level, not at 105 feet below sea level.

Choice D is incorrect. This is the rate by which the pressure, in psi, exerted on the scuba diver increases for each foot the scuba diver descends below sea level.

Question Difficulty:
Medium

Question ID e470e19d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: e470e19d

The function f is defined by $f(x) = 7x - 84$. What is the *x*-intercept of the graph of $y = f(x)$ in the *xy*-plane?

A. $(-12, 0)$

B. $(-7, 0)$

C. $(7, 0)$

D. $(12, 0)$

ID: e470e19d Answer

Correct Answer:
D

Rationale

Choice D is correct. The given function f is a linear function. Therefore, the graph of $y = f(x)$ in the *xy*-plane has one *x*-intercept at the point $(k, 0)$, where k is a constant. Substituting 0 for $f(x)$ and k for x in the given function yields $0 = 7k - 84$. Adding 84 to both sides of this equation yields $84 = 7k$. Dividing both sides of this equation by 7 yields $12 = k$. Therefore, the *x*-intercept of the graph of $y = f(x)$ in the *xy*-plane is $(12, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID af711d1b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: af711d1b

Distance (kilometers)	Average time (minutes)
0.32	8
0.56	14
0.68	17

The table gives the average time t, in minutes, it takes Carly to travel a certain distance d, in kilometers. Which equation could represent this linear relationship?

A. $t = 4d$

B. $t = \frac{1}{25}d$

C. $t = 25d$

D. $t = \frac{1}{4}d$

ID: af711d1b Answer

Correct Answer:

C

Rationale

Choice C is correct. The average time t, in minutes, it takes Carly to travel a certain distance d, in kilometers, is given in the table. This linear relationship can be represented by an equation in the form $t = ad + b$, where a and b are constants. The table shows that it takes Carly an average time of 8 minutes to travel 0.32 kilometers. Substituting 8 for t and 0.32 for d in the equation $t = ad + b$ yields $8 = 0.32a + b$. Subtracting $0.32a$ from both sides of this equation yields $8 - 0.32a = b$. The table also shows that it takes Carly an average time of 14 minutes to travel 0.56 kilometers. Substituting 14 for t and 0.56 for d in the equation $t = ad + b$ yields $14 = 0.56a + b$. Subtracting $0.56a$ from both sides of this equation yields $14 - 0.56a = b$. Substituting $8 - 0.32a$ for b in this equation yields $14 - 0.56a = 8 - 0.32a$. Subtracting 8 from both sides of this equation yields $6 - 0.56a = -0.32a$. Adding $0.56a$ to both sides of this equation yields $6 = 0.24a$. Dividing both sides of this equation by 0.24 yields $25 = a$. Substituting 25 for a in the equation $8 = 0.32a + b$ yields $8 = 0.32(25) + b$, or $8 = 8 + b$. Subtracting 8 from both sides of this equation yields $0 = b$. Substituting 25 for a and 0 for b in the equation $t = ad + b$ yields $t = 25d + 0$, or $t = 25d$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium

Question ID f7e39fe9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: f7e39fe9

x	10	15	20	25
$f(x)$	82	137	192	247

The table shows four values of x and their corresponding values of $f(x)$. There is a linear relationship between x and $f(x)$ that is defined by the equation $f(x) = mx - 28$, where m is a constant. What is the value of m?

ID: f7e39fe9 Answer

Correct Answer:
11

Rationale

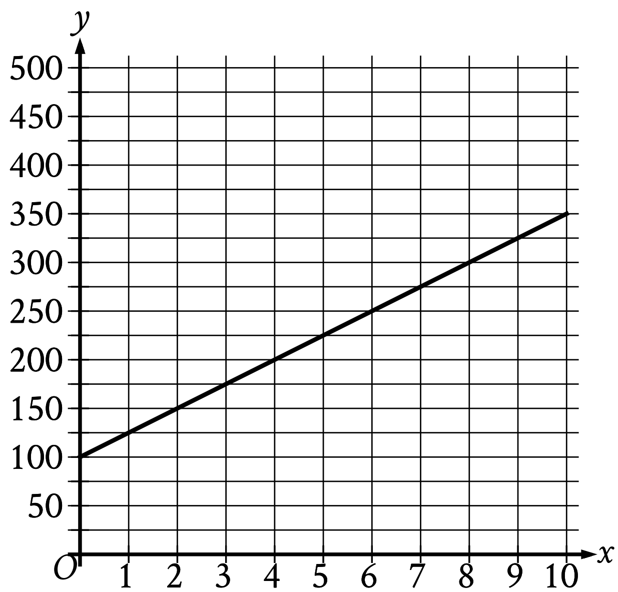
The correct answer is 11. It's given that $f(x)$ is defined by the equation $f(x) = mx - 28$, where m is a constant. It's also given in the table that when $x = 10$, $f(x) = 82$. Substituting 10 for x and 82 for $f(x)$ in the equation $f(x) = mx - 28$ yields, $82 = m(10) - 28$. Adding 28 to both sides of this equation yields $110 = 10m$. Dividing both sides of this equation by 10 yields $11 = m$. Therefore, the value of m is 11.

Question Difficulty:
Medium

Question ID 5cf1bbc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 5cf1bbc9



The graph of the function f, where $y = f(x)$, gives the total cost y, in dollars, for a certain video game system and x games. What is the best interpretation of the slope of the graph in this context?

A. Each game costs $\$25$.

B. The video game system costs $\$100$.

C. The video game system costs $\$25$.

D. Each game costs $\$100$.

ID: 5cf1bbc9 Answer

Correct Answer:

A

Rationale

Choice A is correct. The given graph is a line, and the slope of a line is defined as the change in the value of y for each increase in the value of x by 1. It's given that y represents the total cost, in dollars, and that x represents the number of games. Therefore, the change in the value of y for each increase in the value of x by 1 represents the change in total cost, in dollars, for each increase in the number of games by 1. In other words, the slope represents the cost, in dollars, per game. The graph shows that when the value of x increases from 0 to 1, the value of y increases from 100 to 125. It follows that the slope is 25, or the cost per game is $\$25$. Thus, the best interpretation of the slope of the graph is that each game costs $\$25$.

Choice B is incorrect. This is an interpretation of the *y*-intercept of the graph rather than the slope of the graph.

Choice C is incorrect. The slope of the graph is the cost per game, not the cost of the video game system.

Choice D is incorrect. Each game costs $\$25$, not $\$100$.

Question Difficulty:
Medium

Question ID c81f1c2f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c81f1c2f

$$f(x) = 45x + 600$$

The function f gives the monthly fee $f(x)$, in dollars, a facility charges to keep x crates in storage. What is the monthly fee, in dollars, the facility charges to keep 50 crates in storage?

ID: c81f1c2f Answer

Correct Answer:
2850

Rationale

The correct answer is $2{,}850$. It's given that the function $f(x) = 45x + 600$ gives the monthly fee, in dollars, a facility charges to keep x crates in storage. Substituting 50 for x in this function yields $f(50) = 45(50) + 600$, or $f(50) = 2{,}850$. Therefore, the monthly fee, in dollars, the facility charges to keep 50 crates in storage is $2{,}850$.

Question Difficulty:
Medium

Question ID 0d5495a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0d5495a6

$$h(x) = x + b$$

For the linear function h, b is a constant and $h(0) = 45$. What is the value of b?

ID: 0d5495a6 Answer

Correct Answer:
45

Rationale

The correct answer is 45. It's given that $h(0) = 45$. Therefore, for the given function h, when $x = 0$, $h(x) = 45$. Substituting 0 for x and 45 for $h(x)$ in the given function, $h(x) = x + b$, yields $45 = 0 + b$, or $45 = b$. Therefore, the value of b is 45.

Question Difficulty:
Medium

Question ID 3e9eaffc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3e9eaffc

Caleb used juice to make popsicles. The function $f(x) = -5x + 30$ approximates the volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. Which statement is the best interpretation of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane in this context?

A. Caleb used approximately 5 fluid ounces of juice for each popsicle.

B. Caleb had approximately 5 fluid ounces of juice when he began to make the popsicles.

C. Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

D. Caleb used approximately 30 fluid ounces of juice for each popsicle.

ID: 3e9eaffc Answer

Correct Answer:
C

Rationale

Choice C is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m represents the slope and b represents the *y*-intercept, $(0, b)$, of the line of $y = f(x)$ in the *xy*-plane. The function $f(x) = -5x + 30$ is linear. Therefore, the graph of the given function $y = f(x)$ in the *xy*-plane has a *y*-intercept of $(0, 30)$. It's given that $f(x)$ gives the approximate volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. It follows that the *y*-intercept of $(0, 30)$ means that Caleb had approximately 30 fluid ounces of juice remaining after making 0 popsicles. In other words, Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

Choice A is incorrect. This is an interpretation of the slope, rather than the *y*-intercept, of the graph of $y = f(x)$ in the *xy*-plane.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:
Medium

Question ID a775af14

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a775af14

In the *xy*-plane, the graph of the linear function f contains the points $(0, 2)$ and $(8, 34)$. Which equation defines f, where $y = f(x)$?

A. $f(x) = 2x + 42$

B. $f(x) = 32x + 36$

C. $f(x) = 4x + 2$

D. $f(x) = 8x + 2$

ID: a775af14 Answer

Correct Answer:
C

Rationale

Choice C is correct. In the *xy*-plane, the graph of a linear function can be written in the form $f(x) = mx + b$, where m represents the slope and $(0, b)$ represents the *y*-intercept of the graph of $y = f(x)$. It's given that the graph of the linear function f, where $y = f(x)$, in the *xy*-plane contains the point $(0, 2)$. Thus, $b = 2$. The slope of the graph of a line containing any two points (x_1, y_1) and (x_2, y_2) can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $(0, 2)$ and $(8, 34)$, it follows that the slope of the graph of the line containing these points is $m = \frac{34-2}{8-0}$, or $m = 4$. Substituting 4 for m and 2 for b in $f(x) = mx + b$ yields $f(x) = 4x + 2$.

Choice A is incorrect. This function represents a graph with a slope of 2 and a *y*-intercept of $(0, 42)$.

Choice B is incorrect. This function represents a graph with a slope of 32 and a *y*-intercept of $(0, 36)$.

Choice D is incorrect. This function represents a graph with a slope of 8 and a *y*-intercept of $(0, 2)$.

Question Difficulty:
Medium

Question ID 9551cd91

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 9551cd91

Number of cars	Maximum number of passengers and crew
3	174
5	284
10	559

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

A. $55c - p = -9$

B. $55c - p = 9$

C. $55p - c = -9$

D. $55p - c = 9$

ID: 9551cd91 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that there is a linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. It follows that this relationship can be represented by an equation of the form $p = mc + b$, where m is the rate of change of p in this relationship and b is a constant. The rate of change of p in this relationship can be calculated by dividing the difference in any two values of p by the difference in the corresponding values of c. Using two pairs of values given in the table, the rate of change of p in this relationship is $\frac{284-174}{5-3}$, or 55. Substituting 55 for m in the equation $p = mc + b$ yields $p = 55c + b$. The value of b can be found by substituting any value of c and its corresponding value of p for c and p, respectively, in this equation. Substituting 10 for c and 559 for p yields $559 = 55(10) + b$, or $559 = 550 + b$. Subtracting 550 from both sides of this equation yields $9 = b$. Substituting 9 for b in the equation $p = 55c + b$ yields $p = 55c + 9$. Subtracting 9 from both sides of this equation yields $p - 9 = 55c$. Subtracting p from both sides of this equation yields $-9 = 55c - p$, or $55c - p = -9$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID ebd457f4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: ebd457f4

Brian saves $\frac{2}{5}$ of the $\$215$ he earns each week from his job. If Brian continues to save at this rate, how much money, in dollars, will Brian save in 9 weeks?

ID: ebd457f4 Answer

Correct Answer:
774

Rationale

The correct answer is 774. It's given that Brian saves $\frac{2}{5}$ of the $\$215$ he earns each week from his job. Therefore, Brian saves $\left(\frac{2}{5}\right)(\$215)$, or $\$86$, per week. If Brian continues to save at this rate of $\$86$ per week for 9 weeks, then he will save a total of $(9)(86)$, or 774, dollars.

Question Difficulty:
Medium

Question ID 6c5c2a81

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 6c5c2a81

For the linear function h, the graph of $y = h(x)$ in the *xy*-plane passes through the points $(7, 21)$ and $(9, 25)$. Which equation defines h?

A. $h(x) = \frac{1}{2}x - \frac{7}{2}$

B. $h(x) = 2x + 7$

C. $h(x) = 7x + 21$

D. $h(x) = 9x + 25$

ID: 6c5c2a81 Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that the graph of the linear function h, where $y = h(x)$, passes through the points $(7, 21)$ and $(9, 25)$ in the *xy*-plane. An equation defining h can be written in the form $y = mx + b$, where $y = h(x)$, m represents the slope of the graph in the *xy*-plane, and b represents the *y*-coordinate of the *y*-intercept of the graph. The slope can be found using any two points, (x_1, y_1) and (x_2, y_2), and the formula $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$. Substituting $(7, 21)$ and $(9, 25)$ for (x_1, y_1) and (x_2, y_2), respectively, in the slope formula yields $m = \frac{25-21}{9-7}$, which is equivalent to $m = \frac{4}{2}$, or $m = 2$. Substituting 2 for m and $(7, 21)$ for (x, y) in the equation $y = mx + b$ yields $21 = (2)(7) + b$, or $21 = 14 + b$. Subtracting 14 from each side of this equation yields $7 = b$. Substituting 2 for m and 7 for b in the equation $y = mx + b$ yields $y = 2x + 7$. Since $y = h(x)$, it follows that the equation that defines h is $h(x) = 2x + 7$.

Choice A is incorrect. For this function, the graph of $y = h(x)$ in the *xy*-plane would pass through $(7, 0)$, not $(7, 21)$, and $(9, 1)$, not $(9, 25)$.

Choice C is incorrect. For this function, the graph of $y = h(x)$ in the *xy*-plane would pass through $(7, 70)$, not $(7, 21)$, and $(9, 84)$, not $(9, 25)$.

Choice D is incorrect. For this function, the graph of $y = h(x)$ in the *xy*-plane would pass through $(7, 88)$, not $(7, 21)$, and $(9, 106)$, not $(9, 25)$.

Question Difficulty:

Medium

Question ID dae126d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: dae126d7

The boiling point of water at sea level is 212 degrees Fahrenheit $(°F)$. For every 550 feet above sea level, the boiling point of water is lowered by about $1°F$. Which of the following equations can be used to find the boiling point *B* of water, in $°F$, *x* feet above sea level?

A. $B = 550 + \frac{x}{212}$

B. $B = 550 - \frac{x}{212}$

C. $B = 212 + \frac{x}{550}$

D. $B = 212 - \frac{x}{550}$

ID: dae126d7 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the boiling point of water at sea level is 212°F and that for every 550 feet above sea level, the boiling point of water is lowered by about 1°F. Therefore, the change in the boiling point of water x feet above sea level is represented by the expression $-\frac{x}{550}$. Adding this expression to the boiling point of water at sea level gives the equation for the boiling point B of water, in °F, x feet above sea level: $B = -\frac{x}{550} + 212$, or $B = 212 - \frac{x}{550}$.

Choices A and B are incorrect and may result from using the boiling point of water at sea level as the rate of change and the rate of change as the initial boiling point of water at sea level. Choice C is incorrect and may result from representing the change in the boiling point of water as an increase rather than a decrease.

Question Difficulty:
Medium

Question ID 271f7e3f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 271f7e3f

$$f(x) = \frac{(x+7)}{4}$$

For the function *f* defined above, what is the value of $f(9) - f(1)$?

A. 1

B. 2

C. $\frac{1}{4}$

D. $\frac{9}{4}$

ID: 271f7e3f Answer

Correct Answer:

B

Rationale

Choice B is correct. The value of $f(9) - f(1)$ can be calculated by finding the values of $f(9)$ and $f(1)$. The value of $f(9)$ can be found by substituting 9 for x in the given function: $f(9) = \frac{(9+7)}{4}$. This equation can be rewritten as $f(9) = \frac{16}{4}$, or 4. Then, the value of $f(1)$ can be found by substituting 1 for x in the given function: $f(1) = \frac{(1+7)}{4}$. This equation can be rewritten as $f(1) = \frac{8}{4}$, or 2. Therefore, $f(9) - f(1) = 4 - 2$, which is equivalent to 2.

Choices A, C, and D are incorrect and may result from incorrectly substituting values of x in the given function or making computational errors.

Question Difficulty:

Medium

Question ID c651cc56

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c651cc56

x	$f(x)$
0	−2
2	4
6	16

Some values of the linear function f are shown in the table above. What is the value of $f(3)$?

A. 6

B. 7

C. 8

D. 9

ID: c651cc56 Answer

Correct Answer:
B

Rationale

Choice B is correct. A linear function has a constant rate of change, and any two rows of the table shown can be used to calculate this rate. From the first row to the second, the value of x is increased by 2 and the value of $f(x)$ is increased by $6 = 4 - (-2)$. So the values of $f(x)$ increase by 3 for every increase by 1 in the value of x. Since $f(2) = 4$, it follows that $f(2+1) = 4 + 3 = 7$. Therefore, $f(3) = 7$.

Choice A is incorrect. This is the third x-value in the table, not $f(3)$. Choices C and D are incorrect and may result from errors when calculating the function's rate of change.

Question Difficulty:
Medium

Question ID c22b5f25

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c22b5f25

In the *xy*-plane, the points $(-2,3)$ and $(4,-5)$ lie on the graph of which of the following linear functions?

A. $f(x)=x+5$

B. $f(x)=\frac{1}{2}x+4$

C. $f(x)=-\frac{4}{3}x+\frac{1}{3}$

D. $f(x)=-\frac{3}{2}x+1$

ID: c22b5f25 Answer

Correct Answer:
C

Rationale

Choice C is correct. A linear function can be written in the form $f(x)=mx+b$, where m is the slope and b is the y-coordinate of the y-intercept of the line. The slope of the graph can be found using the formula $m=\frac{y_2-y_1}{x_2-x_1}$. Substituting the values of the given points into this formula yields $m=\frac{-5-3}{4-(-2)}$ or $m=\frac{-8}{6}$, which simplifies to $m=-\frac{4}{3}$. Only choice C shows an equation with this slope.

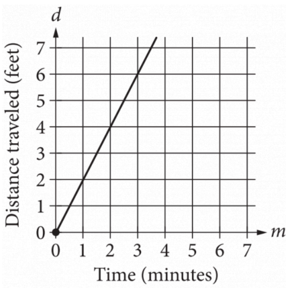
Choices A, B, and D are incorrect and may result from computation errors or misinterpreting the given information.

Question Difficulty:
Medium

Question ID 11e1ab81

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 11e1ab81



The graph above shows the distance traveled *d*, in feet, by a product on a conveyor belt *m* minutes after the product is placed on the belt. Which of the following equations correctly relates *d* and *m* ?

A. $d=2m$

B. $d=\frac{1}{2}m$

C. $d=m+2$

D. $d=2m+2$

ID: 11e1ab81 Answer

Correct Answer:
A

Rationale

Choice A is correct. The line passes through the origin. Therefore, this is a relationship of the form $d=km$, where k is a constant representing the slope of the graph. To find the value of k, choose a point (m,d) on the graph of the line other than the origin and substitute the values of m and d into the equation. For example, if the point $(2,4)$ is chosen, then $4=k(2)$, and $k=2$. Therefore, the equation of the line is $d=2m$.

Choice B is incorrect and may result from calculating the slope of the line as the change in time over the change in distance traveled instead of the change in distance traveled over the change in time. Choices C and D are incorrect because each of these equations represents a line with a d-intercept of 2. However, the graph shows a line with a d-intercept of 0.

Question Difficulty:
Medium

Question ID 4fe4fd7c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 4fe4fd7c

$$c(x) = mx + 500$$

A company's total cost $c(x)$, in dollars, to produce *x* shirts is given by the function above, where *m* is a constant and $x > 0$. The total cost to produce 100 shirts is $800. What is the total cost, in dollars, to produce 1000 shirts? (Disregard the $ sign when gridding your answer.)

ID: 4fe4fd7c Answer

Rationale

The correct answer is 3500. The given information includes a cost, $800, to produce 100 shirts. Substituting $c(x) = 800$ and $x = 100$ into the given equation yields $800 = m \cdot 100 + 500$. Subtracting 500 from both sides of the equation yields $300 = m \cdot 100$. Dividing both sides of this equation by 100 yields $3 = m$. Substituting the value of m into the given equation yields $c(x) = 3x + 500$. Substituting 1000 for x in this equation and solving for $c(x)$ gives the cost of 1000 shirts: $3(1000) + 500$, or 3500.

Question Difficulty:
Medium

Question ID 2f34cd1c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2f34cd1c

The function $f(x) = 55.20 - 0.16x$ gives the estimated surface water temperature $f(x)$, in degrees Celsius, of a body of water on the $x\text{th}$ day of the year, where $220 \le x \le 360$. Based on the model, what is the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year?

A. 55.20

B. 3.04

C. -0.16

D. -52.16

ID: 2f34cd1c Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the function $f(x) = 55.20 - 0.16x$ gives the estimated surface water temperature, in degrees Celsius, of a body of water on the $x\text{th}$ day of the year. Substituting 326 for x in the given function yields $f(326) = 55.20 - 0.16(326)$, which is equivalent to $f(326) = 55.20 - 52.16$, or $f(326) = 3.04$. Therefore, the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year is 3.04.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the rate of change, in degrees Celsius per day, of the estimated surface water temperature.

Choice D is incorrect. This is the change, in degrees Celsius, in the estimated surface water temperature over 326 days.

Question Difficulty:
Medium

Question ID 3122fc7b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3122fc7b

A linear model estimates the population of a city from 1991 to 2015. The model estimates the population was 57 thousand in 1991, 224 thousand in 2011, and x thousand in 2015. To the nearest whole number, what is the value of x?

ID: 3122fc7b Answer

Correct Answer:
257

Rationale

The correct answer is 257. It's given that a linear model estimates the population of a city from 1991 to 2015. Since the population can be estimated using a linear model, it follows that there is a constant rate of change for the model. It's also given that the model estimates the population was 57 thousand in 1991, 224 thousand in 2011, and x thousand in 2015. The change in the population between 2011 and 1991 is $224 - 57$, or 167, thousand. The change in the number of years between 2011 and 1991 is $2011 - 1991$, or 20, years. Dividing 167 by 20 gives $167/20$, or 8.35, thousand per year. Thus, the change in population per year from 1991 to 2015 estimated by the model is 8.35 thousand. The change in the number of years between 2015 and 2011 is $2015 - 2011$, or 4, years. Multiplying the change in population per year by the change in number of years yields the increase in population from 2011 to 2015 estimated by the model: $(8.35)(4)$, or 33.4, thousand. Adding the change in population from 2011 to 2015 estimated by the model to the estimated population in 2011 yields the estimated population in 2015. Thus, the estimated population in 2015 is $33.4 + 224$, or 257.4, thousand. Therefore to the nearest whole number, the value of x is 257.

Question Difficulty:
Medium

Question ID 23dedddd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 23dedddd

In the linear function f, $f(0) = 8$ and $f(1) = 12$. Which equation defines f?

A. $f(x) = 12x + 8$

B. $f(x) = 4x$

C. $f(x) = 4x + 12$

D. $f(x) = 4x + 8$

ID: 23dedddd Answer

Correct Answer:
D

Rationale

Choice D is correct. Since f is a linear function, it can be defined by an equation of the form $f(x) = ax + b$, where a and b are constants. It's given that $f(0) = 8$. Substituting 0 for x and 8 for $f(x)$ in the equation $f(x) = ax + b$ yields $8 = a(0) + b$, or $8 = b$. Substituting 8 for b in the equation $f(x) = ax + b$ yields $f(x) = ax + 8$. It's given that $f(1) = 12$. Substituting 1 for x and 12 for $f(x)$ in the equation $f(x) = ax + 8$ yields $12 = a(1) + 8$, or $12 = a + 8$. Subtracting 8 from both sides of this equation yields $a = 4$. Substituting 4 for a in the equation $f(x) = ax + 8$ yields $f(x) = 4x + 8$. Therefore, an equation that defines f is $f(x) = 4x + 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID c8fb6bcb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c8fb6bcb

$$f(x) = 2x + 244$$

The given function f represents the perimeter, in centimeters (cm), of a rectangle with a length of x cm and a fixed width. What is the width, in cm, of the rectangle?

A. 2

B. 122

C. 244

D. 488

ID: c8fb6bcb Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that $f(x) = 2x + 244$ represents the perimeter, in centimeters (cm), of a rectangle with a length of x cm and a fixed width. If w represents a fixed width, in cm, then the perimeter, in cm, of a rectangle with a length of x cm and a fixed width of w cm can be given by the function $f(x) = 2x + 2w$. Therefore, $2x + 2w = 2x + 244$. Subtracting $2x$ from both sides of this equation yields $2w = 244$. Dividing both sides of this equation by 2 yields $w = 122$. Therefore, the width, in cm, of the rectangle is 122.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID a91a2b75

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a91a2b75

The function f is defined by $f(x) = -9x + 9$. What is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

ID: a91a2b75 Answer

Correct Answer:

9

Rationale

The correct answer is 9. The *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane is the point where the graph of $y = f(x)$ crosses the *y*-axis, which occurs at $x = 0$. It's given that the function f is defined by $f(x) = -9x + 9$. Substituting 0 for x and y for $f(x)$ in this equation yields $y = -9(0) + 9$, or $y = 9$. It follows that $y = 9$ when $x = 0$ and that the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane is $(0, 9)$. Therefore, the *y*-coordinate of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane is 9.

Question Difficulty:

Medium

Question ID 56dc8045

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 56dc8045

$$w(t) = 300 - 4t$$

The function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. According to the model, what is the predicted volume, in milliliters, draining from the container each second?

A. 300

B. 296

C. 75

D. 4

ID: 56dc8045 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. The given function $w(t) = 300 - 4t$ can be rewritten as $w(t) = -4t + 300$. Thus, for each increase of t by 1, the value of $w(t)$ decreases by $4(1)$, or 4. Therefore, the predicted volume, in milliliters, draining from the container each second is 4 milliliters.

Choice A is incorrect. This is the amount of liquid, in milliliters, in the container before the liquid begins draining.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:
Medium

Question ID c01f4a95

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c01f4a95

$$j(x) = mx + 144$$

For the linear function j, m is a constant and $j(12) = 18$. What is the value of $j(10)$?

ID: c01f4a95 Answer

Correct Answer:

39

Rationale

The correct answer is 39. It's given that for the linear function j, m is a constant and $j(12) = 18$. Substituting 12 for x and 18 for $j(x)$ in the given equation yields $18 = m(12) + 144$. Subtracting 144 from both sides of this equation yields $-126 = m(12)$. Dividing both sides of this equation by 12 yields $-10.5 = m$. Substituting -10.5 for m in the given equation, $j(x) = mx + 144$, yields $j(x) = -10.5x + 144$. Substituting 10 for x in this equation yields $j(10) = (-10.5)(10) + 144$, or $j(10) = 39$. Therefore, the value of $j(10)$ is 39.

Question Difficulty:

Medium

Question ID 4b0c156b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 4b0c156b

In the linear function h, $h(28) = 15$ and $h(26) = 22$. Which equation defines h?

A. $h(x) = -\frac{2}{7}x + 23$

B. $h(x) = -\frac{2}{7}x + 113$

C. $h(x) = -\frac{7}{2}x + 23$

D. $h(x) = -\frac{7}{2}x + 113$

ID: 4b0c156b Answer

Correct Answer:

D

Rationale

Choice D is correct. An equation defining h can be written in the form $y = mx + b$, where $y = h(x)$, m represents the slope of the graph of $y = h(x)$ in the *xy*-plane, and b represents the *y*-coordinate of the *y*-intercept of the graph. It's given that $h(28) = 15$ and $h(26) = 22$. It follows that the points $(28, 15)$ and $(26, 22)$ are on the graph of $y = h(x)$ in the *xy*-plane. The slope can be found by using any two points, (x_1, y_1) and (x_2, y_2), and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(28, 15)$ and $(26, 22)$ for (x_1, y_1) and (x_2, y_2), respectively, in the slope formula yields $m = \frac{22-15}{26-28}$, or $m = -\frac{7}{2}$. Substituting $-\frac{7}{2}$ for m and $(28, 15)$ for (x, y) in the equation $y = mx + b$ yields $15 = \left(-\frac{7}{2}\right)(28) + b$, or $15 = -98 + b$. Adding 98 to both sides of this equation yields $113 = b$. Substituting $-\frac{7}{2}$ for m and 113 for b in the equation $y = mx + b$ yields $y = -\frac{7}{2}x + 113$. Since $y = h(x)$, it follows that the equation that defines h is $h(x) = -\frac{7}{2}x + 113$.

Choice A is incorrect. For this function, $h(26) = \frac{109}{7}$, not $h(26) = 22$.

Choice B is incorrect. For this function, $h(28) = 105$, not $h(28) = 15$, and $h(26) = \frac{739}{7}$, not $h(26) = 22$.

Choice C is incorrect. For this function, $h(28) = -75$, not $h(28) = 15$, and $h(26) = -68$, not $h(26) = 22$.

Question Difficulty:

Medium

Question ID 868fc236

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 868fc236

Energy per Gram of Typical Macronutrients

Macronutrient	Food calories	Kilojoules
Protein	4.0	16.7
Fat	9.0	37.7
Carbohydrate	4.0	16.7

The table above gives the typical amounts of energy per gram, expressed in both food calories and kilojoules, of the three macronutrients in food. If x food calories is equivalent to k kilojoules, of the following, which best represents the relationship between x and k ?

A. $k = 0.24x$

B. $k = 4.2x$

C. $x = 4.2k$

D. $xk = 4.2$

ID: 868fc236 Answer

Correct Answer:

B

Rationale

Choice B is correct. The relationship between x food calories and k kilojoules can be modeled as a proportional relationship. Let (x_1, k_1) and (x_2, k_2) represent the values in the first two rows in the table: $(4.0, 16.7)$ and $(9.0, 37.7)$. The rate of change, or $\frac{(k_2 - k_1)}{(x_2 - x_1)}$, is $\frac{21}{5} = 4.2$; therefore, the equation that best represents the relationship between x and k is $k = 4.2x$.

Choice A is incorrect and may be the result of calculating the rate of change using $\frac{(x_2 - x_1)}{(k_2 - k_1)}$. Choice C is incorrect because the number of kilojoules is greater than the number of food calories. Choice D is incorrect and may be the result of an error when setting up the equation.

Question Difficulty:

Medium

Question ID 7e1bff94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 7e1bff94

The table gives the number of hours, h, of labor and a plumber's total charge $f(h)$, in dollars, for two different jobs.

h	$f(h)$
1	155
3	285

There is a linear relationship between h and $f(h)$. Which equation represents this relationship?

A. $f(h) = 25h + 130$

B. $f(h) = 130h + 25$

C. $f(h) = 65h + 90$

D. $f(h) = 90h + 65$

ID: 7e1bff94 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that there is a linear relationship between a plumber's hours of labor, h, and the plumber's total charge $f(h)$, in dollars. It follows that the relationship can be represented by an equation of the form $f(h) = mh + b$, where m is the rate of change of the function f and b is a constant. The rate of change of f can be calculated by dividing the difference in two values of $f(h)$ by the difference in the corresponding values of h. Based on the values given in the table, the rate of change of f is $\frac{285-155}{3-1}$, or 65. Substituting 65 for m in the equation $f(h) = mh + b$ yields $f(h) = 65h + b$. The value of b can be found by substituting any value of h and its corresponding value of $f(h)$ for h and $f(h)$, respectively, in this equation. Substituting 1 for h and 155 for $f(h)$ yields $155 = 65(1) + b$, or $155 = 65 + b$. Subtracting 65 from both sides of this equation yields $90 = b$. Substituting 90 for b in the equation $f(h) = 65h + b$ yields $f(h) = 65h + 90$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 946ab892

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 946ab892

For the linear function g, the graph of $y = g(x)$ in the *xy*-plane has a slope of 2 and passes through the point $(1, 14)$. Which equation defines g?

A. $g(x) = 2x$

B. $g(x) = 2x + 2$

C. $g(x) = 2x + 12$

D. $g(x) = 2x + 14$

ID: 946ab892 Answer

Correct Answer:
C

Rationale

Choice C is correct. An equation defining a linear function can be written in the form $g(x) = mx + b$, where m is the slope and $(0, b)$ is the *y*-intercept of the graph of $y = g(x)$ in the *xy*-plane. It's given that the graph of $y = g(x)$ has a slope of 2. Therefore, $m = 2$. It's also given that the graph of $y = g(x)$ passes through the point $(1, 14)$. It follows that when $x = 1$, $g(x) = 14$. Substituting 1 for x, 14 for $g(x)$, and 2 for m in the equation $g(x) = mx + b$ yields $14 = 2(1) + b$, or $14 = 2 + b$. Subtracting 2 from each side of this equation yields $12 = b$. Therefore, $b = 12$. Substituting 2 for m and 12 for b in the equation $g(x) = mx + b$ yields $g(x) = 2x + 12$. Therefore, the equation that defines g is $g(x) = 2x + 12$.

Choice A is incorrect. For this function, the graph of $y = g(x)$ in the *xy*-plane passes through the point $(1, 2)$, not $(1, 14)$.

Choice B is incorrect. For this function, the graph of $y = g(x)$ in the *xy*-plane passes through the point $(1, 4)$, not $(1, 14)$.

Choice D is incorrect. For this function, the graph of $y = g(x)$ in the *xy*-plane passes through the point $(1, 16)$, not $(1, 14)$.

Question Difficulty:
Medium

Question ID 0cadb20e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0cadb20e

The function f is defined by $f(x) = \frac{x+15}{5}$, and $f(a) = 10$, where a is a constant. What is the value of a?

A. 5

B. 10

C. 35

D. 65

ID: 0cadb20e Answer

Correct Answer:

C

Rationale

Choice C is correct. It's given that $f(x) = \frac{x+15}{5}$ and $f(a) = 10$, where a is a constant. Therefore, for the given function f, when $x = a$, $f(x) = 10$. Substituting a for x and 10 for $f(x)$ in the given function f yields $10 = \frac{a+15}{5}$. Multiplying both sides of this equation by 5 yields $50 = a + 15$. Subtracting 15 from both sides of this equation yields $35 = a$. Therefore, the value of a is 35.

Choice A is incorrect. This is the value of a if $f(a) = 4$.

Choice B is incorrect. This is the value of a if $f(a) = 5$.

Choice D is incorrect. This is the value of a if $f(a) = 16$.

Question Difficulty:

Medium

Question ID 042aa429

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 042aa429

If $f(x) = x + 7$ and $g(x) = 7x$, what is the value of $4f(2) - g(2)$?

A. -5

B. 1

C. 22

D. 28

ID: 042aa429 Answer

Correct Answer:
C

Rationale

Choice C is correct. The value of $f(2)$ can be found by substituting 2 for x in the given equation $f(x) = x + 7$, which yields $f(2) = 2 + 7$, or $f(2) = 9$. The value of $g(2)$ can be found by substituting 2 for x in the given equation $g(x) = 7x$, which yields $g(2) = 7(2)$, or $g(2) = 14$. The value of the expression $4f(2) - g(2)$ can be found by substituting the corresponding values into the expression, which gives $4(9) - 14$. This expression is equivalent to $36 - 14$, or 22.

Choice A is incorrect. This is the value of $f(2) - g(2)$, not $4f(2) - g(2)$.

Choice B is incorrect and may result from calculating $4f(2)$ as $4(2) + 7$, rather than $4(2 + 7)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID b2fe7ab6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: b2fe7ab6

x	$g(x)$
1	54
2	51
3	48
4	45

For the linear function g, the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b?

A. 3

B. 27

C. 54

D. 57

ID: b2fe7ab6 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that for the linear function g, the table shows four values of x and their corresponding values of $g(x)$. It's also given that the function can be written as $g(x) = mx + b$, where m and b are constants. The table shows that when the value of x is 1, the corresponding value of $g(x)$ is 54. Substituting 1 for x and 54 for $g(x)$ in $g(x) = mx + b$ yields $54 = m(1) + b$ or, $54 = m + b$. Subtracting b from both sides of this equation yields $54 - b = m$. The table also shows that when the value of x is 2, the corresponding value of $g(x)$ is 51. Substituting 2 for x and 51 for $g(x)$ in $g(x) = mx + b$ yields $51 = m(2) + b$, or $51 = 2m + b$. Substituting $54 - b$ for m in this equation yields $51 = 2(54 - b) + b$. Applying the distributive property to the right-hand side of this equation yields $51 = 108 - 2b + b$, or $51 = 108 - b$. Subtracting 108 from both sides of this equation yields $-57 = -b$. Dividing both sides of this equation by -1 yields $57 = b$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 1bc11c4e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1bc11c4e

$$g(m) = -0.05m + 12.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

A. 0.05

B. 12.1

C. 20

D. 242.0

ID: 1bc11c4e Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that the function g models the number of gallons that remain from a full gas tank in a car after driving m miles. In the given function $g(m) = -0.05m + 12.1$, the coefficient of m is -0.05. This means that for every increase in the value of m by 1, the value of $g(m)$ decreases by 0.05. It follows that for each mile driven, there is a decrease of 0.05 gallons of gasoline. Therefore, 0.05 gallons of gasoline are used to drive each mile.

Choice B is incorrect and represents the number of gallons of gasoline in a full gas tank.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:
Medium

Question ID 113b938e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 113b938e

$y = 18 - 5x$

The equation above represents the speed y, in feet per second, of Sheila's bicycle x seconds after she applied the brakes at the end of a ride. If the equation is graphed in the xy-plane, which of the following is the best interpretation of the x-coordinate of the line's x-intercept in the context of the problem?

A. The speed of Sheila's bicycle, in feet per second, before Sheila applied the brakes

B. The number of feet per second the speed of Sheila's bicycle decreased each second after Sheila applied the brakes

C. The number of seconds it took from the time Sheila began applying the brakes until the bicycle came to a complete stop

D. The number of feet Sheila's bicycle traveled from the time she began applying the brakes until the bicycle came to a complete stop

ID: 113b938e Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that for each point (x, y) on the graph of the given equation, the x-coordinate represents the number of seconds after Sheila applied the brakes, and the y-coordinate represents the speed of Sheila's bicycle at that moment in time. For the graph of the equation, the y-coordinate of the x-intercept is 0. Therefore, the x-coordinate of the x-intercept of the graph of the given equation represents the number of seconds it took from the time Sheila began applying the brakes until the bicycle came to a complete stop.

Choice A is incorrect. The speed of Sheila's bicycle before she applied the brakes is represented by the y-coordinate of the y-intercept of the graph of the given equation, not the x-coordinate of the x-intercept. Choice B is incorrect. The number of feet per second the speed of Sheila's bicycle decreased each second after Sheila applied the brakes is represented by the slope of the graph of the given equation, not the x-coordinate of the x-intercept. Choice D is incorrect and may result from misinterpreting x as the distance, in feet, traveled after applying the brakes, rather than the time, in seconds, after applying the brakes.

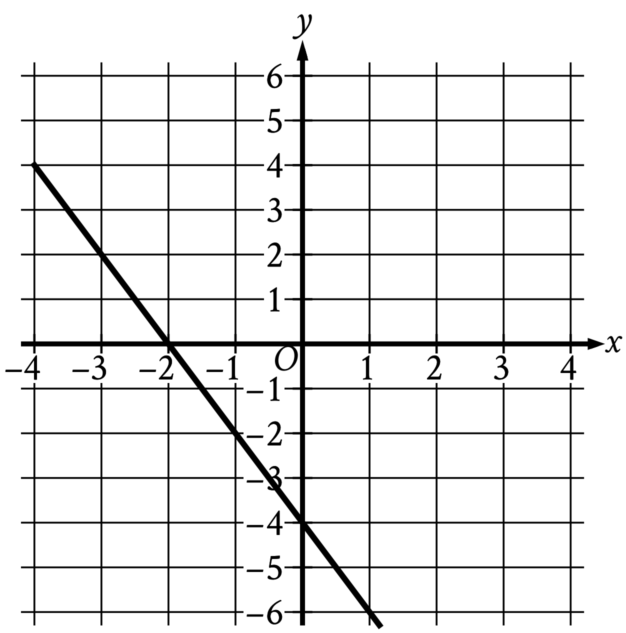
Question Difficulty:
Medium

Question ID 295a41f0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 295a41f0

The graph of $y = f(x) - 11$ is shown.



Which equation defines the linear function f?

A. $f(x) = -13x - 11$

B. $f(x) = -2x + 7$

C. $f(x) = -13x + 7$

D. $f(x) = -2x - 11$

ID: 295a41f0 Answer

Correct Answer:

B

Rationale

Choice B is correct. The graph of a line in the *xy*-plane can be represented by the equation $y = mx + b$, where m is the slope of the line and $(0, b)$ is the *y*-intercept. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $(-1, -2)$ and $(0, -4)$. Substituting $(-1, -2)$ and $(0, 4)$ for (x_1, y_1) and (x_2, y_2), respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{-4-(-2)}{0-(-1)}$, which is equivalent to $m = \frac{-2}{1}$, or $m = -2$. Since the line shown passes through the point $(0, -4)$, it follows that $b = -4$. Substituting -2 for m and -4 for b in the equation $y = mx + b$ yields $y = -2x - 4$. It's given that the graph shown is the graph of $y = f(x) - 11$. Substituting $-2x - 4$ for y in the equation $y = f(x) - 11$ yields $-2x - 4 = f(x) - 11$. Adding 11 to both sides of this equation yields $-2x + 7 = f(x)$. Therefore, the equation $f(x) = -2x + 7$ defines the linear function f.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 1f0966db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1f0966db

The function f is defined by $f(x) = \frac{9}{7}x + \frac{8}{7}$. For what value of x does $f(x) = 5$?

ID: 1f0966db Answer

Correct Answer:

3

Rationale

The correct answer is 3. Substituting 5 for $f(x)$ in the given function yields $5 = \frac{9}{7}x + \frac{8}{7}$. Multiplying each side of this equation by 7 yields $7(5) = 7\left(\frac{9}{7}x + \frac{8}{7}\right)$, or $35 = 9x + 8$. Subtracting 8 from each side of this equation yields $27 = 9x$. Dividing each side of this equation by 9 yields $3 = x$. Therefore, $f(x) = 5$ when the value of x is 3.

Question Difficulty:

Medium

Question ID 441558e7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 441558e7

Scientists collected fallen acorns that each housed a colony of the ant species *P. ohioensis* and analyzed each colony's structure. For any of these colonies, if the colony has x worker ants, the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y, in the colony. If one of these colonies has 58 worker ants, which of the following is closest to the predicted number of larvae in the colony?

A. 41

B. 61

C. 83

D. 190

ID: 441558e7 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y, in a colony of ants if the colony has x worker ants. If one of these colonies has 58 worker ants, the predicted number of larvae in that colony can be found by substituting 58 for x in the given equation. Substituting 58 for x in the given equation yields $y = 0.67(58) + 2.6$, or $y = 41.46$. Of the given choices, 41 is closest to the predicted number of larvae in the colony.

Choice B is incorrect. This is closest to the predicted number of larvae in a colony with 87 worker ants.

Choice C is incorrect. This is closest to the number of worker ants for which the predicted number of larvae in a colony is 58.

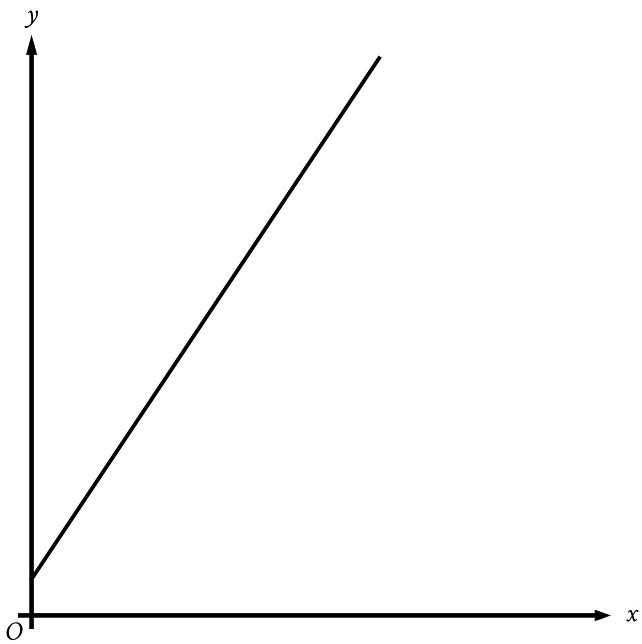
Choice D is incorrect. This is closest to the predicted number of larvae in a colony with 280 worker ants.

Question Difficulty:
Medium

Question ID f0773a55

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: f0773a55



The graph represents the total charge, in dollars, by an electrician for x hours of work. The electrician charges a onetime fee plus an hourly rate. What is the best interpretation of the slope of the graph?

A. The electrician’s hourly rate

B. The electrician’s onetime fee

C. The maximum amount that the electrician charges

D. The total amount that the electrician charges

ID: f0773a55 Answer

Correct Answer:

A

Rationale

Choice A is correct. It’s given that the electrician charges a onetime fee plus an hourly rate. It's also given that the graph represents the total charge, in dollars, for x hours of work. This graph shows a linear relationship in the *xy*-plane. Thus, the total charge y, in dollars, for x hours of work can be represented as $y = mx + b$, where m is the slope and $(0, b)$ is the *y*-intercept of the graph of the equation in the *xy*-plane. Since the given graph represents the total charge, in dollars, by an electrician for x hours of work, it follows that its slope is m, or the electrician’s hourly rate.

Choice B is incorrect. The electrician's onetime fee is represented by the *y*-coordinate of the *y*-intercept, not the slope, of the graph.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:
Medium

Question ID 53487897

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 53487897

The relationship between two variables, x and y, is linear. For every increase in the value of x by 1, the value of y increases by 8. When the value of x is 2, the value of y is 18. Which equation represents this relationship?

A. $y = 2x + 18$

B. $y = 2x + 8$

C. $y = 8x + 2$

D. $y = 3x + 26$

ID: 53487897 Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that the relationship between x and y is linear. An equation representing a linear relationship can be written in the form $y = mx + b$, where m is the slope and b is the *y*-coordinate of the *y*-intercept of the graph of the relationship in the *xy*-plane. It’s given that for every increase in the value of x by 1, the value of y increases by 8. The slope of a line can be expressed as the change in y over the change in x. Thus, the slope, m, of the line representing this relationship can be expressed as $\frac{8}{1}$, or 8. Substituting 8 for m in the equation $y = mx + b$ yields $y = 8x + b$. It's also given that when the value of x is 2, the value of y is 18. Substituting 2 for x and 18 for y in the equation $y = 8x + b$ yields $18 = 8(2) + b$, or $18 = 16 + b$. Subtracting 16 from each side of this equation yields $2 = b$. Substituting 2 for b in the equation $y = 8x + b$ yields $y = 8x + 2$. Therefore, the equation $y = 8x + 2$ represents this relationship.

Choice A is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 2, not 8, and when the value of x is 2, the value of y is 22, not 18.

Choice B is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 2, not 8, and when the value of x is 2, the value of y is 12, not 18.

Choice D is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 3, not 8, and when the value of x is 2, the value of y is 32, not 18.

Question Difficulty:
Medium

Question ID 3ce92ce8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3ce92ce8

$$f(x) = 2x + 3$$

For the given function f, the graph of $y = f(x)$ in the *xy*-plane is parallel to line j. What is the slope of line j?

ID: 3ce92ce8 Answer

Correct Answer:

2

Rationale

The correct answer is 2. It's given that function f is defined by $f(x) = 2x + 3$. Therefore, the equation representing the graph of $y = f(x)$ in the *xy*-plane is $y = 2x + 3$, and the graph is a line. For a linear equation in the form $y = mx + b$, m represents the slope of the line. Since the value of m in the equation $y = 2x + 3$ is 2, the slope of the line defined by function f is 2. It's given that line j is parallel to the line defined by function f. The slopes of parallel lines are equal. Therefore, the slope of line j is also 2.

Question Difficulty:

Medium

Question ID 8a6de407

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 8a6de407

The function f is defined by $f(x) = mx + b$, where m and b are constants. If $f(0) = 18$ and $f(1) = 20$, what is the value of m ?

ID: 8a6de407 Answer

Rationale

The correct answer is 2. The slope-intercept form of an equation for a line is $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. Two ordered pairs, (x_1, y_1) and (x_2, y_2), can be used to compute the slope using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that $f(0) = 18$ and $f(1) = 20$; therefore, the two ordered pairs for this line are $(0, 18)$ and $(1, 20)$. Substituting these values for (x_1, y_1) and (x_2, y_2) gives $\frac{20 - 18}{1 - 0} = \frac{2}{1}$, or 2.

Question Difficulty:
Medium

Question ID 41fdc0b8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 41fdc0b8

Population of Greenleaf, Idaho

Year	Population
2000	862
2010	846

The table above shows the population of Greenleaf, Idaho, for the years 2000 and 2010. If the relationship between population and year is linear, which of the following functions P models the population of Greenleaf t years after 2000?

A. $P(t) = 862 - 1.6t$

B. $P(t) = 862 - 16t$

C. $P(t) = 862 + 16(t - 2{,}000)$

D. $P(t) = 862 - 1.6(t - 2{,}000)$

ID: 41fdc0b8 Answer

Correct Answer:
A

Rationale

Choice A is correct. It is given that the relationship between population and year is linear; therefore, the function that models the population t years after 2000 is of the form $P(t) = mt + b$, where m is the slope and b is the population when $t = 0$. In the year 2000, $t = 0$. Therefore, $b = 862$. The slope is given by $m = \frac{P(10) - P(0)}{10 - 0} = \frac{846 - 862}{10 - 0} = \frac{-16}{10} = -1.6$. Therefore, $P(t) = -1.6t + 862$, which is equivalent to the equation in choice A.

Choice B is incorrect and may be the result of incorrectly calculating the slope as just the change in the value of P. Choice C is incorrect and may be the result of the same error as in choice B, in addition to incorrectly using t to represent the year, instead of the number of years after 2000. Choice D is incorrect and may be the result of incorrectly using t to represent the year instead of the number of years after 2000.

Question Difficulty:
Medium

Question ID 8ad64841

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 8ad64841

A linear function f gives a company's profit, in dollars, for selling x items. The company's profit is $\$220$ when it sells 8 items, and its profit is $\$320$ when it sells 10 items. Which equation defines f?

A. $f(x) = 150x - 320$

B. $f(x) = 32x$

C. $f(x) = 50x - 10x$

D. $f(x) = 50x - 180$

ID: 8ad64841 Answer

Correct Answer:

D

Rationale

Choice D is correct. It's given that the relationship between x and $f(x)$ is linear. A linear function can be written in the form $f(x) = mx + b$, where m is the slope and b is the *y*-coordinate of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane. Given two points on a line, (x_1, y_1) and (x_2, y_2), the slope of the line can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that the company's profit is $\$220$ when it sells 8 items and the profit is $\$320$ when it sells 10 items. Since $f(x)$ represents the company's profit, in dollars, for selling x items, the graph of $y = f(x)$ in the *xy*-plane passes through the points $(8, 220)$ and $(10, 320)$. Substituting $(8, 220)$ and $(10, 320)$ for (x_1, y_1) and (x_2, y_2), respectively, in the slope formula yields $m = \frac{320 - 220}{10 - 8}$, which gives $m = \frac{100}{2}$, or $m = 50$. Substituting 50 for m, 8 for x, and 220 for $f(x)$ in $f(x) = mx + b$ yields $220 = (50)(8) + b$, or $220 = 400 + b$. Subtracting 400 from each side of this equation yields $-180 = b$. Substituting 50 for m and -180 for b in $f(x) = mx + b$ yields $f(x) = 50x - 180$. Therefore, the equation that defines f is $f(x) = 50x - 180$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium